

CCR Null-Regime Conditional Closure v31

QICN Internal Hardening Artifact

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Definition 1 (Typed perturbation boundary). *For a system S with identity object $\mathcal{I}(S)$, let $\Omega_{\text{int}}(S)$ denote finite-energy perturbations whose domain and codomain remain inside the admissible identity structure of S . Let $\text{Ext}(S)$ denote extension-side witnesses that compare $\mathcal{I}(S)$ with an independently specified object $\tilde{\mathcal{I}}$ not introduced as an element of $\Omega_{\text{int}}(S)$.*

Definition 2 (Undefined-assignment marker and separated external witness). *The symbol $\perp$ (“bottom”) denotes absence of an admissible phenomenological assignment. It is disjoint from $\mathcal{E}$, so $\perp \notin \mathcal{E}$. This is distinct from the null regime $\emptyset_\phi \in \mathcal{E}$, which is the bottom element of the phenomenological poset in Paper 3. A separated external witness for a phenomenological assignment Φ is a tuple $(\tilde{\mathcal{I}}, \epsilon, C)$ such that:*

1. $\tilde{\mathcal{I}} \in \text{Ext}(S)$ and $d_{\mathcal{I}}(\mathcal{I}(S), \tilde{\mathcal{I}}) = \epsilon > 0$;
2. $C > 0$ is a lower Lipschitz constant on the declared two-point witness domain; and
3. Φ is defined and admissible on both $\mathcal{I}(S)$ and $\tilde{\mathcal{I}}$, so $\Phi(\mathcal{I}(S)), \Phi(\tilde{\mathcal{I}}) \in \mathcal{E}$.

Theorem 1 (Witness-relative undefined-assignment exclusion). *Let S be a CCR channel with $M_\Omega = +\infty$. Let Φ be a declared compatibility operator satisfying the lower Lipschitz condition on the separated witness domain. Suppose a separated external witness $(\tilde{\mathcal{I}}, \epsilon, C)$ exists for Φ . Then Φ cannot be undefined on both witness endpoints:*

$$\neg(\Phi(\mathcal{I}(S)) = \perp \wedge \Phi(\tilde{\mathcal{I}}) = \perp).$$

Proof. Suppose for contradiction that both $\Phi(\mathcal{I}(S)) = \perp$ and $\Phi(\tilde{\mathcal{I}}) = \perp$. By the definition of $\perp$, neither endpoint has an admissible $\mathcal{E}$ -valued phenomenological assignment. This contradicts the separated-witness hypothesis, which requires Φ to be defined and admissible on both endpoints, so that $\Phi(\mathcal{I}(S)), \Phi(\tilde{\mathcal{I}}) \in \mathcal{E}$ and the lower Lipschitz condition is meaningful there.

The case where both endpoints are assigned the null regime $\emptyset_\phi \in \mathcal{E}$ is a different typed statement: it is governed by the Paper 3 instability theorem and uses the metric on $\mathcal{E}$. The present theorem concerns only the absence marker $\perp \notin \mathcal{E}$. $\square$

Remark 1 (Non-claim). *CCR alone does not produce a witness, does not prove that a compatibility operator exists, and does not imply consciousness or experience. The theorem is conditional on a typed external witness and a checked lower Lipschitz bound. In particular, $\perp$ is not a point in $\mathcal{E}$; it marks absence of assignment. The null regime $\emptyset_\phi \in \mathcal{E}$ is a separate concept: it is the bottom element of the phenomenological poset, not the absence of a value.*